

MuRoDance: Music-Anticipatory Humanoid Dance through Discrete–Continuous Motion Generation

Abstract—Humanoid dance to arriving music requires anticipating musical events while continuing coherently from motion already committed. We present MuRoDance, a framework for streaming music-driven humanoid dance that combines predicted musical context with robot-native discrete–continuous (D+C) motion generation. A frozen music foundation model forecasts future music states from heard audio alone. These states are aligned with upcoming motion times and condition both discrete structural planning and continuous residual generation. Kinematics-guided reconstruction objectives assign explicit structural responsibilities to discrete codes, while their combination with continuous residuals reconstructs full motion through a shared decoder. Commit Forcing trains continuation from jointly committed structure and detail and the reference boundary state reconstructed from their decoded motion. Committed references are delivered to a fixed whole-body controller, whose measured-state feedback remains separate from the generator’s reference-context update. On the evaluated 60-second FineDance generations, a frozen music–motion evaluator favors the intended soundtrack over alternative tracks in approximately 75% of pairwise comparisons under the reported evaluation and selection protocol. Matched history-training comparisons reveal a tradeoff between motion distribution fidelity and smoothness. These results characterize musical correspondence and motion quality in generated robot-native references.

I. INTRODUCTION

Dance preparation begins before a visible accent: weight shifts precede a step, and the body transitions toward a pose before reaching it. A humanoid dancing to arriving music must prepare its movements before hearing all the sound they will accompany. Waiting for an event leaves less time to prepare; reading the future recording changes the task into offline generation. Streaming further requires each new movement to continue from emitted motion that can no longer be revised. Musical anticipation and continuation from committed motion are therefore coupled challenges for music-driven humanoid dance.

Music-conditioned choreography has advanced through expressive motion models and large paired datasets [1]–[4]. DiscoForcing [5] extends causal audio-driven generation to streaming motion and humanoid deployment, while RoboPerform [6] connects audio-derived style with expressive humanoid control. Building on these advances, we study how a humanoid can continuously generate dance that is appropriate to arriving music. Music does not prescribe a unique choreography, but its rhythm, accents, and dynamic variation should be reflected in coherent movement and remain perceptible after execution. This motivates a coupled question: how can predicted musical context guide upcoming robot-native motion while preserving continuity with the

movement already committed?

We present Music-Anticipatory Humanoid Dance through Discrete–Continuous Motion Generation (MuRoDance), a framework for streaming music-driven humanoid dance that uses predicted musical context to guide upcoming movement. A frozen Magenta RealTime 2 (MRT-2) music foundation model [7] forecasts future music states from the audio received so far (Fig. 1(a)), providing musical lookahead without access to future waveform samples. Music pretraining supplies the continuation prior, while paired dance data teaches the robot-native motion generator to interpret its predictions. Aligned with upcoming motion times, these forecast states condition both discrete structural planning and continuous-detail generation, connecting musical anticipation to whole-body organization and local articulation.

Expressing musical conditions through a humanoid requires coordinated whole-body structure, expressive local articulation, and continuity across successive motion decisions. Our kinematics-guided discrete–continuous (D+C) representation couples discrete structure and continuous detail through complementary reconstruction objectives (Fig. 1(c)). Structure-only reconstruction supervises designated body organization and endpoint placement, while the combined latent reconstructs full motion through a shared boundary-conditioned decoder. Building on hybrid motion representations [8], this supervision assigns explicit kinematic responsibilities to the discrete component while retaining complementary variation in the full latent.

During streaming generation, both the motion history and the next boundary state depend on the model’s own output rather than recorded motion. Commit Forcing trains continuation after jointly sampling and committing both components (Fig. 1(d)). Their decoded movement determines the next reference boundary state, pairing generated history with the state it actually induces. The next plan is trained using this generated history–state pair.

At each decision, MuRoDance plans a short motion segment, commits a prefix, and updates its reference context for the next plan (Fig. 1(b)). A reference interface delivers the committed motion to a fixed SONIC whole-body controller [9]. The generator continues from reference-derived history and boundary state, while measured robot state closes the controller’s physical feedback loop. Because reference delivery and tracking can alter timing and motion dynamics, we formulate a paired evaluation protocol to examine music–motion matching, music adaptation, and dance quality in generated references and executed trajectories.

Our contributions are:

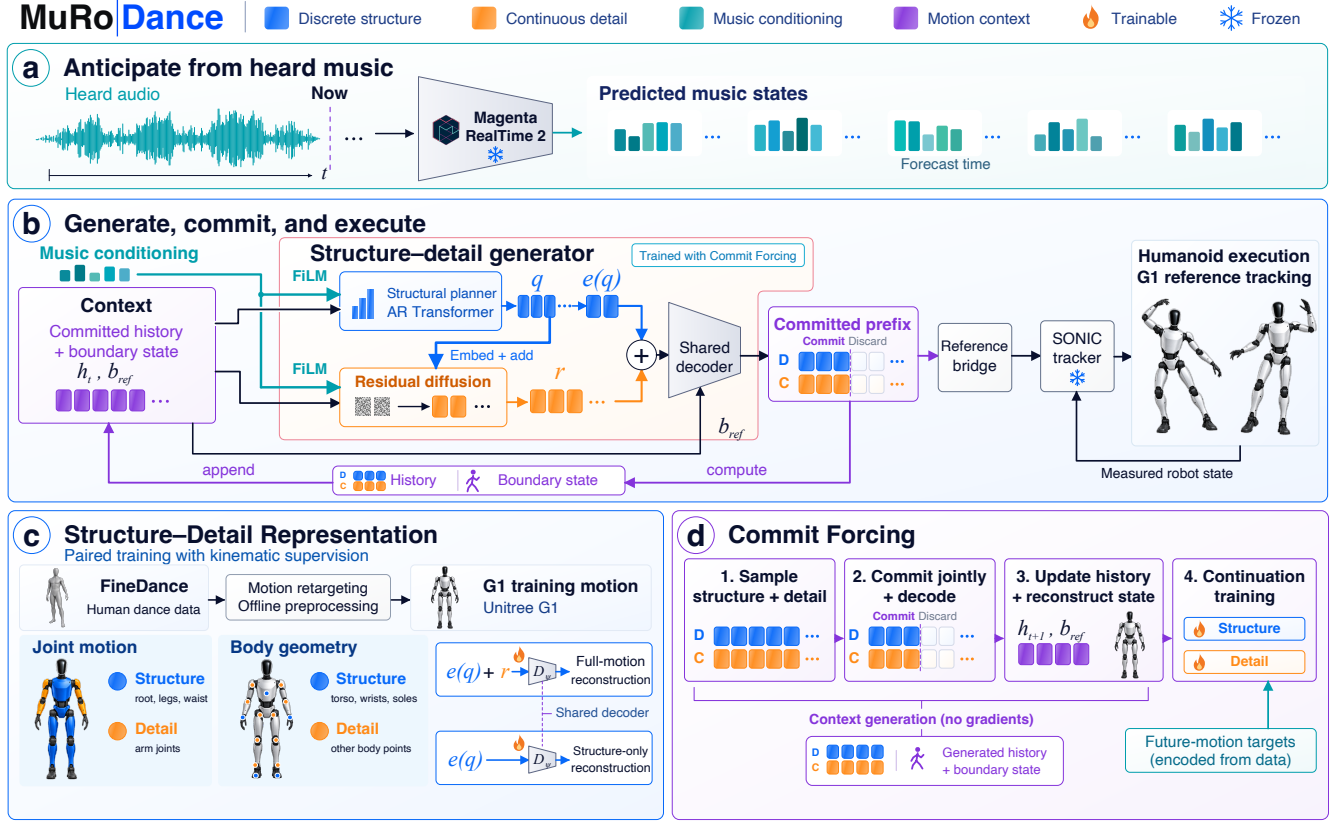


Fig. 1. **Overview of MuRoDance.** (a) Frozen MRT-2 forecasts upcoming music states from arrived audio. (b) Forecast conditions modulate the discrete planner and residual diffusion model through FiLM; sampled structure also conditions residual generation. A shared decoder produces robot-native references, whose committed prefix is delivered through a bridge to fixed SONIC control. Committed latent codes update the generator history, and decoded motion determines its next reference boundary state. Measured robot state feeds back only to the controller. (c) Retargeted FineDance motion supports D+C learning through full-motion and structure-only reconstruction with a shared decoder and complementary kinematic supervision. (d) Commit Forcing constructs a generated history–state pair under stop-gradient and trains continuation using ground-truth future-motion targets. Robot poses illustrate the method and are not hardware results.

- **Anticipatory music conditioning for humanoid dance.** Future music states predicted from arrived audio provide temporally aligned conditions for discrete structural planning and continuous-detail generation.
- **Kinematics-guided robot-native D+C generation.** Complementary reconstruction objectives assign explicit structural responsibilities to discrete codes, whose embeddings combine with continuous residuals through a shared boundary-conditioned decoder.
- **Commit Forcing for streaming continuation.** Joint commitment of generated structure and detail constructs a consistent history–state pair, enabling continuation training from the sampled reference transition.

Our experimental protocol separates robot-native motion quality from musical adaptation, assessing rhythm, tempo, annotated style, and dynamic expression through complementary measures and learned correspondence. Music-condition interventions test whether generated dance changes appropriately with its soundtrack; matched ablations examine musical forecasting, D+C supervision, and Commit Forcing. Paired simulation comparisons distinguish properties of committed generation from changes introduced by reference delivery

and control. Section V discusses the interpretation and scope of these measurements.

II. RELATED WORK

A. Learning-Based Humanoid Control and Motion Tracking

Reinforcement learning and motion imitation provide foundations for expressive whole-body control. Neural Categorical Priors [10] and Generative Pretrained Controllers [11] connect learned motion vocabularies with physics-based character control. Reference-conditioned policies further enable reusable motion tracking: PHC [12] demonstrates tracking and recovery in simulated avatars, while ExBody2 [13], BeyondMimic [14], and SONIC [9] advance expressive and versatile humanoid control. These methods establish how reference movement can be realized under physical dynamics. Our work uses a pretrained whole-body controller as the execution layer and focuses on generating the music-conditioned references it follows. The controller retains measured-state feedback, while the generator maintains its own committed reference context.

B. Music-Conditioned Motion Generation

Music-to-dance systems use explicit acoustic descriptors, such as those extracted with librosa [15], or learned audio representations from models including Jukebox [16], MERT [17], and MuQ [18]. AIST++/FACT [1] and Bailando [19] develop music-conditioned choreography through sequence modeling and discrete motion representations. EDGE [2] and LODGE [20] advance diffusion-based and long-form dance generation, while FineDance [3], MEGADance [21], and SoulNet [4] address fine-grained motion, genre-aware generation, and music–motion alignment.

Streaming generation additionally constrains the audio available before each motion decision. DiscoForcing [5] uses causal audio conditioning and diffusion forcing for streaming motion and humanoid deployment. Our music interface supplies explicit forecasts of upcoming music states from the observed prefix. Generative music models, including Live Music Models [22] and MRT-2 [7], provide the continuation prior used for this purpose. We investigate how their predicted states can condition upcoming robot motion without access to future waveform samples. The music forecaster remains frozen; paired dance training learns how to use its predictions.

C. Motion Representations and Generation

Discrete motion models such as T2M-GPT [23] and MoMask [24] use quantized vocabularies, while MLD [25] generates motion through continuous latent diffusion. DiscoRD [26] connects discrete tokens with continuous decoding. Most closely, DC-Motion [8] combines discrete structure and continuous residuals. Our D+C design builds on this hybrid principle through robot-kinematic reconstruction objectives that assign designated structural responsibilities to discrete codes, with full motion reconstructed through a shared boundary-conditioned decoder.

Recursive generation also introduces a mismatch between recorded training context and generated inference context. Scheduled Sampling [27] and Diffusion Forcing [28] address related sequence-training problems. MotionStreamer [29] combines causal motion compression with diffusion-based autoregression and Two-Forward training. Commit Forcing focuses on the transition induced by our D+C generator: sampled structure and detail enter the committed history together, and their decoded motion determines the next reference boundary state. Its distinguishing operation is the joint construction of this history–state pair for continuation training.

D. Integrating Motion Generation with Humanoid Control

Connecting conditional generation to physical control enables humanoids to express higher-level intent. HAR-MON [30] generates whole-body humanoid motion from language descriptions. RoboPerform [6] combines motion content with audio-derived style through a teacher–student control framework, while DiscoForcing [5] connects streaming motion generation to humanoid execution through retargeting and tracking. These systems establish complementary

interfaces between external conditions, generated movement, and robot control.

Other approaches use physical execution to inform generation or adapt the controller. PhysDiff [31] incorporates physics-based correction into denoising, and PARC [32] uses physically tracked motion to augment generator training. BeyondMimic [14] extends tracking to guided generative motion composition, while terrain-aware humanoid generation [33] couples robot-native references with tracker adaptation. Our framework performs retargeting during data preparation, generates robot-native references online, and delivers committed motion to fixed SONIC control. Continuation training operates within the generated reference process. We therefore distinguish reference generation, delivery, and measured execution, and assess musical correspondence and motion quality alongside tracking accuracy.

III. METHOD

MuRoDance couples three decisions: what music to anticipate, how to express the response as structured motion, and how to continue after committing it. Figure 1 shows the complete system; the following figures expand the reconstruction and continuation mechanisms.

a) Streaming task: At decision time τ , $a_{\leq\tau}$ is the heard audio, h is committed code history, and b is the pose and velocity state reconstructed from the last committed reference frames. A frozen music model predicts $m^{\text{pred}} = F(a_{\leq\tau})$, and the generator samples

$$(q_{1:H}, r_{1:H}) \sim p_{\theta}(q, r \mid h, b, m^{\text{pred}}). \quad (1)$$

Here q is discrete structure and r is continuous detail. Their sum decodes to native G1 motion. Both streams commit the same first $C < H$ steps; the remaining plan is discarded. We use $H = 8$ and $C = 4$ at 15 latent steps per second, yielding 16 planned and eight committed frames at 30 Hz.

A. Anticipatory Music Conditioning

SpectroStream tokenizes heard audio, and frozen MRT-2 Small rolls forward to predict musical states [7]. Its music pretraining supplies the continuation prior; paired dance training learns how to turn that prior into movement. Fourteen forecast states at 25 Hz cover approximately 0.56 s, alongside the 0.53 s motion plan. Future waveform samples are never used as input.

a) Align time before modulating features: Forecast and motion steps have different rates, so matching array indices would assign the wrong musical time. For state u_j with physical lead ℓ_j relative to the decision, we form

$$c_j = \text{WLN}(u_j) + \phi(\ell_j). \quad (2)$$

Here W projects the music features and ϕ encodes their lead time. We linearly interpolate neighboring projected states at motion lead $d_i = i/15$ to obtain $\hat{c}_i$. A coverage mask m_i is one only within the valid forecast support and zero when the condition is absent or no longer covers that motion time.

Each conditioned block adds a gated, frame-wise modulation [34]:

$$v'_i = v_i + m_i g \odot [s(\hat{c}_i) \odot \text{LN}(v_i) + t(\hat{c}_i)]. \quad (3)$$

The learned maps s, t predict scale and shift, and g is a learned channel gate. When $m_i = 0$, the block retains its original activation exactly. Separate modulators condition the structural Transformer blocks and residual diffusion heads using the same physical-time alignment. The generator and these mappings are trained; MRT-2 remains frozen.

B. Structure–Detail Representation and Paired Training

An encoder maps native motion and its starting state to $z = E_\varphi(x, b)$. Quantization selects $q = Q(z)$ with codebook embedding $e(q)$, and a continuous residual retains the remaining information:

$$z = e(q) + r. \quad (4)$$

Both components are decoded by one boundary-conditioned decoder $D_\psi(\cdot, b)$. Their roles are learned through complementary reconstruction objectives, rather than separate anatomical decoders.

a) Learning complementary roles: We train two reconstructions with the same decoder and starting state:

$$x_f = D_\psi(e(q) + r, b), \quad x_s = D_\psi(e(q), b). \quad (5)$$

The full path preserves complete movement, while the structure-only path gives the discrete anchor an explicit kinematic responsibility (Fig. 1(c)). Codebook and commitment losses train the quantized representation; Appendix ?? specifies the gradient routes.

The full loss supervises native coordinates, native velocities, forward-kinematics positions, and geometry velocities. Each block is standardized with training-set statistics and receives equal weight. The structural loss uses the corresponding selected coordinates: root, legs, and waist in the native views, and torso, wrists, and soles in the geometry views. Consequently, wrist placement can be structural while arm-joint articulation remains part of the complementary native view. Auxiliary structural reconstruction under residual corruption and a boundary-transition loss reinforce these objectives; their settings and exact partitions remain in Appendix ??.

b) Coupled motion generation: A causal Transformer samples q . Its sampled IDs are embedded and added to the residual model’s input features; a diffusion generator [35] samples r using this structure, motion history, boundary state, and aligned music. The codebook embedding and sampled residual then sum at the shared decoder to produce 38-dimensional native motion. Joint commitment preserves the pairing of the two streams across planning steps.

C. Commit Forcing

Teacher Forcing supplies ground-truth motion history and its boundary state; streaming inference supplies both from the model’s earlier output. Commit Forcing changes this conditioning transition while retaining ground-truth supervision for the subsequent motion (Fig. 3).

a) Construct the next history and state together: A first pass samples $\tilde{q}, \tilde{r}$ with the inference samplers under stop-gradient. Their common committed prefix updates the two parts of the next context:

$$(\tilde{h}^+, \tilde{b}^+) = U(h, b, \tilde{q}_{1:C}, \tilde{r}_{1:C}). \quad (6)$$

U appends the committed codes to history and reconstructs the boundary state from their decoded reference motion. Context replacement selects this entire pair together, so generated history is accompanied by the state of the same commitment. Each supplied training example constructs one such transition independently; the first pass is not a recursive long rollout.

b) Keep the target, change its residual coordinates: The ground-truth structural target q^* and motion latent $z^* = e(q^*) + r^*$ remain fixed. The next structure model is supervised against q^* , while its sampled structure $\hat{q}$ conditions residual denoising. We express the residual target relative to that sampled embedding:

$$r_{\text{rebased}}^* = z^* - e(\hat{q}), \quad e(\hat{q}) + r_{\text{rebased}}^* = z^*. \quad (7)$$

Thus the structure and residual losses retain a common ground-truth target latent despite a changed structural sample. This is latent rebasing, not target re-encoding under $\tilde{b}^+$: the decoder also depends on its boundary state, so latent equality does not imply identical decoded motion.

c) History regularization: After context construction, full-history corruption (FHC) perturbs discrete embeddings and residuals across the motion history. It preserves boundary state, music, masks, and timestamps, and is disabled during inference. Appendix ?? specifies this separate training regularizer.

D. Robot Reference Integration and Execution

As shown in Fig. 1(b), committed robot-native motion is delivered to a pretrained SONIC whole-body controller [9]. A reference bridge reconstructs root motion in the fixed sequence coordinate system, maps joints to the controller convention, and resamples the 30 Hz reference onto a continuous 50 Hz grid. Only the committed prefix is delivered; the remaining prediction is discarded before replanning.

SONIC remains frozen and combines the consumed reference with measured robot state to produce control actions. The generator updates its latent history from the committed discrete codes and continuous residuals, and reconstructs its next boundary state from the corresponding decoded motion. Thus, reference-derived context supports subsequent generation, while measured-state feedback closes the controller’s physical execution loop.

IV. EXPERIMENTS

We evaluate whether MuRoDance generates coherent robot-native dance that responds appropriately to music, and whether these properties are retained during execution. The evaluation covers motion quality and musical adaptation, using motion-distribution metrics, explicit beat alignment,

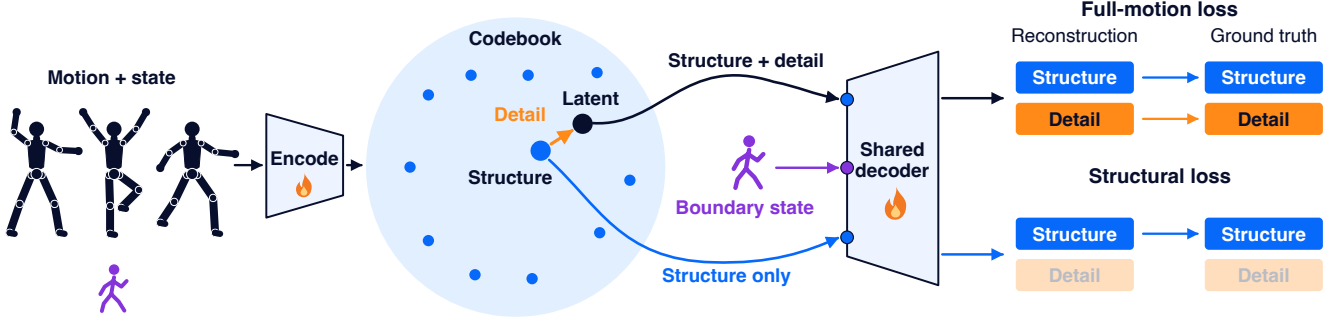


Fig. 2. **Learning complementary structure and detail.** The codebook anchor and continuous residual form the motion latent. Both paths share one state-conditioned decoder and reconstruct complete motion. Blue and orange bands denote structural and complementary kinematic coordinates, including velocities. Arrows indicate comparison with ground truth: the full-motion loss constrains both subsets; the structural loss constrains only blue. Faint orange coordinates remain in the output. Band widths are schematic; flames mark learned modules.

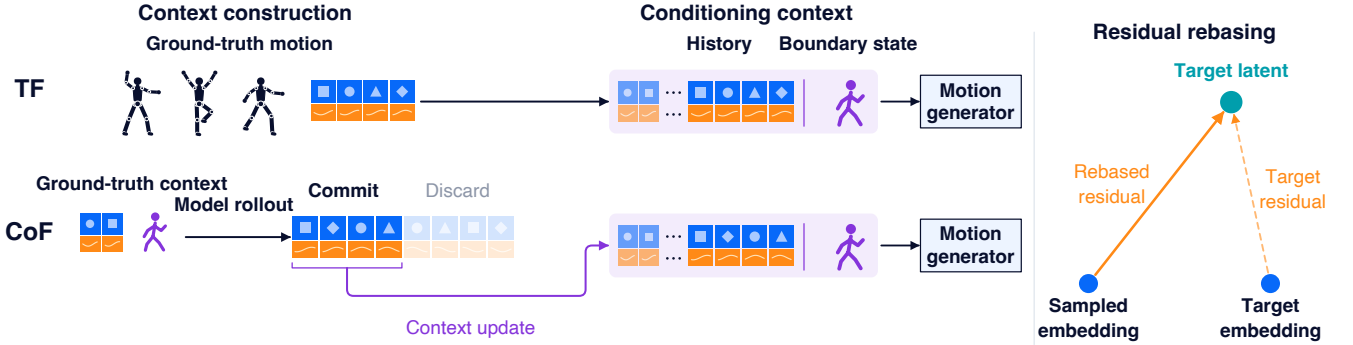


Fig. 3. **Constructing generation contexts with Commit Forcing.** TF uses ground-truth history and its boundary state. CoF samples a motion plan conditioned on ground-truth context, appends its committed prefix to history, and derives the paired boundary state from the decoded committed motion. The discarded tail does not enter the next context. At right, residual rebasing retains the ground-truth motion latent while replacing the target codebook embedding with the sampled embedding as the residual’s origin. Tile counts and geometry are schematic.

and learned music–motion correspondence. Controlled ablations examine musical anticipation, kinematics-guided D+C supervision, and Commit Forcing. Streaming and execution experiments assess timing, reliability, and changes from generated references to tracked motion.

A. Setup and Evaluation Protocol

We use paired FineDance music and human motion re-targeted to the Unitree G1. Data are partitioned by music content into training, validation, and test sets. The generator uses the native 38D representation defined in the method, while evaluation features are extracted from decoded robot motion.

Motion-quality comparisons use 23 s segments from 18 tracks. Long-horizon correspondence and continuation experiments use 60 s segments from eleven tracks. These cohorts are reported separately. Each generator configuration uses three independent training seeds and three sampling seeds per track. Methods share evaluation tracks, initial reference contexts, soundtrack offsets, generation durations, and sampling settings within each comparison. During free-running generation, subsequent contexts evolve from each model’s

own committed outputs. We score the concatenated committed motion using a fixed configuration and initialization rule across metrics.

Evaluator training and model selection exclude the music content used for generator evaluation. All scores use common valid audio–motion intervals and the recorded soundtrack timing. We report training-seed variation and 95% confidence intervals from song-level bootstrap resampling. Paired comparisons resample the same songs across conditions. Scalar scores are aggregated within songs before averaging across songs and training seeds. Distributional metrics are computed from pooled feature sets and recomputed during resampling.

B. Evaluation Metrics

a) *Motion quality:* We report kinetic and geometric Fréchet distances, FID_k and FID_g , between generated and re-targeted-reference motion distributions. Robot-specific kinetic features summarize FK-point speeds and accelerations; geometric features describe root-relative body configurations and inter-point distances. Both populations use the same G1 model and feature extractor. Div_k and Div_g measure diversity in these spaces and are interpreted jointly with distributional

fidelity. PFC provides a foot-motion and root-acceleration-based physical-consistency proxy. Motion activity, jerk, and position/velocity jumps at commitment boundaries provide additional diagnostics of freezing, smoothness, and continuation.

b) Explicit beat alignment: Audio beats are extracted with a fixed librosa beat tracker. Motion beats are local minima of smoothed mean FK-point speed. We report both motion-to-audio and audio-to-motion Beat Alignment Scores (BAS), using a Gaussian matching scale of 0.10s. The two directions measure proximity from the respective event sets. Beat F1 complements BAS through one-to-one event matching with a 0.20s tolerance. Detectors and matching parameters remain fixed across methods.

Dynamic response is additionally measured by the zero-lag Spearman correlation between audio RMS energy and root-relative FK-point speed, aggregated in one-second bins. We interpret this energy-activity association alongside motion quality and retargeted-reference responses.

c) Learned music-motion correspondence: Following SoulNet [4], we adapt a music-motion representation-learning evaluator to robot-native motion. The music branch receives Jukebox features [16]; the motion branch receives root, body-pose, velocity, and contact features. Variational encoders and a shared motion decoder are trained with reconstruction, latent alignment, KL regularization, and contrastive objectives. The evaluator and training-derived normalization are frozen before generator comparison.

For each five-second window, the score s averages audio-motion embedding cosine similarities from three independently trained evaluator instances. For motion x_i , intended music a_i , alternative music a_j , and window w , we define

$$\delta_{ijw} = s(x_i^{(w)}, a_i^{(w)}) - s(x_i^{(w)}, a_j^{(w)}). \quad (8)$$

We report the mean correspondence margin and pairwise accuracy, assigning half credit to ties. Each 60s trajectory is evaluated with five-second windows at a one-second stride against its intended soundtrack and ten fixed alternatives. Window-level margins and decisions are averaged over windows and alternatives before aggregation across sampling repeats, songs, and generator seeds. The gallery and soundtrack offsets are shared across methods.

d) Evaluator validation: Before scoring generated motion, we evaluate the frozen evaluator on held-out retargeted-reference pairs. Table I reports discrimination against ordinary and musically similar alternative tracks. Aligned-versus-shifted audio and original-versus-corrupted motion comparisons further assess temporal sensitivity and susceptibility to degenerate motion. These checks establish the scope in which correspondence scores are interpreted.

C. Motion Quality and Musical Adaptation

a) Overall generation quality: Table II compares MuRoDance with the selected baselines under the common generation protocol. We assess distributional fidelity, diversity, and physical consistency together. Motion activity and qualitative sequences help distinguish smooth dance from

TABLE I
MUSIC-MOTION EVALUATOR VALIDATION. ACCURACY MEASURES
 PREFERENCE FOR THE INTENDED, ALIGNED, OR ORIGINAL PAIR UNDER
 EACH CONTROL. CHANCE ORDERING IS 50%.

Reference-pair comparison	Accuracy (%)	Margin
Ordinary alternative music	—	—
Similar alternative music	—	—
Temporally shifted audio	—	—
Corrupted motion	—	—

TABLE II
MOTION QUALITY ON THE 18-TRACK, 23 S COHORT. LOWER FID AND
 PFC ARE FAVORABLE. DIVERSITY IS INTERPRETED JOINTLY WITH
 FIDELITY. RESULTS AGGREGATE THREE GENERATOR TRAINING SEEDS.

Method	FID _k	FID _g	Div _k	Div _g	PFC
Baseline	—	—	—	—	—
MuRoDance	—	—	—	—	—

inactive or repetitive outputs. For the 18-track cohort, each training seed contributes 54 generated descriptors, compared with 18 reference descriptors under identical crops and extraction settings.

b) Conditional musical adaptation: We generate motion under correct, null, mismatched-track, and temporally shifted music conditions. The initial reference context and sampling settings are paired, while each rollout subsequently evolves independently. All outputs are scored against the predefined target soundtrack. Table III reports rhythmic alignment and learned correspondence for these interventions, with motion quality monitored alongside musical scores.

For paired tracks, we additionally generate one dance per soundtrack and compare intended and swapped music-motion pairings. Musically similar alternatives provide a harder comparison than unrelated tracks. Together, these experiments test whether conditioning changes produce appropriately music-specific movement.

D. Component Ablations

a) Musical anticipation: We compare separately trained heard-only and forecast-conditioned generators with matched motion architectures, training budgets, tracks, and sampling settings. A real-future condition provides an additional reference comparison. Motion quality, beat alignment, and learned correspondence are evaluated over the same intervals. These comparisons isolate the benefit of predictive musical context from the effect of music conditioning itself.

b) Kinematics-guided D+C generation: We compare codecs trained with and without the proposed structural supervision under matched downstream generator training. Full-motion and structure-only reconstruction errors assess the designated kinematic responsibilities of discrete codes. Fixed-structure residual interventions examine complementary variation. Downstream generation metrics test whether this representation-level supervision improves dance quality and musical correspondence.

TABLE III

MUSICAL ADAPTATION UNDER CONDITION INTERVENTION. RESULTS USE THE ELEVEN-TRACK, 60 S COHORT AND FIXED TARGET SOUNDTRACKS. BAS AND F1 ASSESS RHYTHMIC ALIGNMENT; CORRESPONDENCE ACCURACY AND MARGIN ASSESS SOUNDTRACK PREFERENCE.

Music condition	BAS _{m→a}	BAS _{a→m}	Beat F1	MMR accuracy (%)	Margin
Correct	—	—	—	—	—
Null	—	—	—	—	—
Mismatched track	—	—	—	—	—
Temporally shifted	—	—	—	—	—

c) Committed continuation: We compare Teacher Forcing and Commit Forcing on matched 60 s free-running generations. A generated-history-only control retains the recorded boundary state, isolating the effect of constructing the generated history and its resulting state together. We evaluate the full sequence and three consecutive 20 s intervals, reporting musical correspondence, motion activity, and commitment-boundary discontinuities. Clean-history, history-dropout, and full-history-corruption comparisons separately characterize the contribution of history regularization.

E. Streaming and Robot Execution

We evaluate reference execution in MuJoCo using a fixed SONIC whole-body controller. Retargeted reference motion and generated motion pass through the same interface. We distinguish committed generation, delivered reference, controller-consumed target, and measured execution. This separation identifies changes introduced by reference delivery and physical tracking.

a) Streaming performance: Cached and live music conditioning are reported separately, as are saved-trajectory replay and online reference delivery. We measure generation latency, deadline-miss rate, effective future-music coverage at action decisions, and motion-reference preview at controller consumption. All measurements use recorded timestamps. Hardware, planning horizon, commitment length, and sampling settings are fixed within each comparison.

b) Preservation under execution: We compare motion quality and musical adaptation before and after execution using the same soundtrack timeline and common valid intervals. Learned correspondence on executed motion is accompanied by execution-domain evaluator checks. Completion rate, joint tracking error, and amplitude and squared-velocity retention characterize execution reliability and dynamic attenuation. Any lag-compensated tracking analysis is reported separately from audio-synchronized musical scores. Failed runs remain included in completion statistics.

Matched visualizations show how generated dance changes across music inputs and how those movements are retained during tracking. Rendered references and measured execution are identified separately.

V. DISCUSSION

a) Anticipation for embodied musical response: Music-driven humanoid dance requires preparation: weight transfer and coordinated articulation develop before a visible accent. Forecast music states provide prospective context for

these decisions using arrived audio alone. Our framework connects a pretrained musical continuation prior to robot-native motion through paired dance learning. The relevant quantity is the useful musical lead at commitment, which depends jointly on forecast content, temporal alignment, and processing delay. This connects musical anticipation to the timing requirements of embodied movement.

b) Structured expression across commitments: Kinematics-guided D+C supervision assigns discrete codes responsibility for designated whole-body organization while continuous residuals retain complementary variation. These roles are defined through reconstruction objectives and expressed through a shared decoder, allowing structure and detail to remain coupled. Commit Forcing extends this coupling across planning cycles: the committed components determine both the latent history and the reference boundary state used for continuation. This formulation connects the representation of movement to the consequences of previous generation decisions, making their consistency an explicit part of training.

c) Preserving expression through execution: A robot’s musical response depends on how generated movement survives reference delivery and physical tracking. The robot-native interface connects our planner to fixed whole-body control, while paired evaluation examines musical correspondence and motion quality alongside tracking accuracy. This makes the preservation of expressive movement a system-level objective. The current generator plans from committed references, with measured-state feedback handled by SONIC. Extending the planning context to include execution deviations could enable adaptation to the robot’s actual posture and support state. Forecast uncertainty also motivates adapting planning and commitment horizons when musical transitions become less predictable.

VI. CONCLUSION

We presented MuRoDance, a framework for streaming music-driven humanoid dance that integrates musical anticipation, robot-native D+C generation, and committed continuation. A frozen music foundation model forecasts upcoming musical context from arrived audio. Kinematics-guided supervision organizes discrete structure and complementary continuous detail, while Commit Forcing trains the generator to continue from jointly committed motion and its resulting reference state. The generated stream is delivered to a fixed whole-body controller for humanoid execution.

By connecting musical foresight to structured motion and successive commitments, MuRoDance provides a unified approach to generating humanoid dance in response to music as it unfolds.

REFERENCES

- [1] R. Li, S. Yang, D. A. Ross, *et al.*, “AI Choreographer: Music conditioned 3D dance generation with AIST++,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 13 401–13 412.
- [2] J. Tseng, R. Castellon, and C. K. Liu, “EDGE: Editable dance generation from music,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 448–458.
- [3] R. Li, J. Zhao, Y. Zhang, *et al.*, “FineDance: A fine-grained choreography dataset for 3D full body dance generation,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 10 234–10 243.
- [4] X. Li, R. Li, S. Fang, *et al.*, “Music-aligned holistic 3D dance generation via hierarchical motion modeling,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 14 420–14 430. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2025/html/Li_Music-Aligned_Holistic_3D_Dance_Generation_via_Hierarchical_Motion_Modeling_ICCV_2025_paper.html
- [5] K. Ji, B. Qian, B. Wu, *et al.*, “DiscoForcing: A unified framework for real-time audio-driven character control with diffusion forcing,” *arXiv preprint arXiv:2605.28491*, 2026. [Online]. Available: <https://arxiv.org/abs/2605.28491>
- [6] Z. Li, C. Chi, Y. Wei, *et al.*, “Do you have freestyle? expressive humanoid locomotion via audio control,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 956–965. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2026/papers/Li_Do_You_Have_Freestyle_Expressive_Humanoid_Locomotion_via_Audio_Control_CVPR_2026_paper.pdf
- [7] Magenta Team, “Magenta RealTime 2: Open & local live music models,” Magenta blog, June 2026. [Online]. Available: <https://magenta.withgoogle.com/magenta-realtime-2>
- [8] H. Wang, X. Chen, J. Zhang, *et al.*, “DC-Motion: Decoupling structure and details via discrete–continuous tokens for human motion generation,” *arXiv preprint arXiv:2606.14721*, 2026. [Online]. Available: <https://arxiv.org/abs/2606.14721>
- [9] Z. Luo, Y. Yuan, T. Wang, *et al.*, “SONIC: Supersizing motion tracking for natural humanoid whole-body control,” *Science Robotics*, vol. 11, no. 117, p. ead4592, 2026. [Online]. Available: <https://arxiv.org/abs/2511.07820>
- [10] Q. Zhu, H. Zhang, M. Lan, *et al.*, “Neural categorical priors for physics-based character control,” *ACM Transactions on Graphics*, vol. 42, no. 6, 2023.
- [11] Y. Shi, Y. Jiang, C. Tessler, *et al.*, “GPC: Large-scale generative pretraining for transferable motor control,” in *ACM SIGGRAPH 2026 Conference Papers*, 2026.
- [12] Z. Luo, J. Cao, A. Winkler, *et al.*, “Perpetual humanoid control for real-time simulated avatars,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 10 895–10 904. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2023/papers/Luo_Perpetual_Humanoid_Control_for_Real-time_Simulated_Avatars_ICCV_2023_paper.pdf
- [13] M. Ji, X. Peng, F. Liu, *et al.*, “ExBody2: Advanced expressive humanoid whole-body control,” *arXiv preprint arXiv:2412.13196*, 2024. [Online]. Available: <https://arxiv.org/abs/2412.13196>
- [14] Q. Liao, T. E. Truong, X. Huang, *et al.*, “BeyondMimic: From motion tracking to versatile humanoid control via guided diffusion,” *arXiv preprint arXiv:2508.08241*, 2025. [Online]. Available: <https://arxiv.org/abs/2508.08241>
- [15] B. McFee, C. Raffel, D. Liang, *et al.*, “librosa: Audio and music signal analysis in Python,” in *Proceedings of the 14th Python in Science Conference*, 2015, pp. 18–25. [Online]. Available: <https://proceedings.scipy.org/articles/Majora-7b98e3ed-003>
- [16] P. Dhariwal, H. Jun, C. Payne, *et al.*, “Jukebox: A generative model for music,” *arXiv preprint arXiv:2005.00341*, 2020. [Online]. Available: <https://arxiv.org/abs/2005.00341>
- [17] Y. Li, R. Yuan, G. Zhang, *et al.*, “MERT: Acoustic music understanding model with large-scale self-supervised training,” in *International Conference on Learning Representations*, 2024. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2024/hash/33dffa2e3d2ab74a783d1a8c292f66d9-Abstract-Conference.html
- [18] H. Zhu, Y. Zhou, H. Chen, *et al.*, “MuQ: Self-supervised music representation learning with mel residual vector quantization,” *arXiv preprint arXiv:2501.01108*, 2025. [Online]. Available: <https://arxiv.org/abs/2501.01108>
- [19] L. Siyao, W. Yu, T. Gu, *et al.*, “Bailando: 3D dance generation by actor-critic GPT with choreographic memory,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 11 050–11 059.
- [20] R. Li, Y. Zhang, Y. Zhang, *et al.*, “LODGE: A coarse to fine diffusion network for long dance generation guided by the characteristic dance primitives,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 1524–1534.
- [21] K. Yang, X. Tang, Z. Peng, *et al.*, “MEGADance: Mixture-of-experts architecture for genre-aware 3D dance generation,” in *Advances in Neural Information Processing Systems*, vol. 38, 2025. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2025/hash/102bd3c1076dfe98043b19340ac44e08-Abstract-Conference.html
- [22] Lyria Team, A. Caillon, B. McWilliams, *et al.*, “Live music models,” *arXiv preprint arXiv:2508.04651*, 2025. [Online]. Available: <https://arxiv.org/abs/2508.04651>
- [23] J. Zhang, Y. Zhang, X. Cun, *et al.*, “Generating human motion from textual descriptions with discrete representations,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 14 730–14 740.
- [24] C. Guo, Y. Mu, M. G. Javed, *et al.*, “MoMask: Generative masked modeling of 3D human motions,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 1900–1910.
- [25] X. Chen, B. Jiang, W. Liu, *et al.*, “Executing your commands via motion diffusion in latent space,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 18 000–18 010.
- [26] J. Cho, J. Kim, J. Kim, *et al.*, “DisCoRD: Discrete tokens to continuous motion via rectified flow decoding,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 14 602–14 612.
- [27] S. Bengio, O. Vinyals, N. Jaitly, *et al.*, “Scheduled sampling for sequence prediction with recurrent neural networks,” in *Advances in Neural Information Processing Systems*, vol. 28, 2015, pp. 1171–1179.
- [28] B. Chen, D. Martí Monsó, Y. Du, *et al.*, “Diffusion forcing: Next-token prediction meets full-sequence diffusion,” in *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 24 081–24 125.
- [29] L. Xiao, S. Lu, H. Pi, *et al.*, “MotionStreamer: Streaming motion generation via diffusion-based autoregressive model in causal latent space,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 10 086–10 096.
- [30] Z. Jiang, Y. Xie, J. Li, *et al.*, “Harmon: Whole-body motion generation of humanoid robots from language descriptions,” in *Proceedings of the 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2025, pp. 3015–3026. [Online]. Available: <https://proceedings.mlr.press/v270/jiang25b.html>
- [31] Y. Yuan, J. Song, U. Iqbal, *et al.*, “PhysDiff: Physics-guided human motion diffusion model,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 16 010–16 021. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2023/html/Yuan_PhysDiff_Physics-Guided_Human_Motion_Diffusion_Model_ICCV_2023_paper.html
- [32] M. Xu, Y. Shi, K. Yin, *et al.*, “PARC: Physics-based augmentation with reinforcement learning for character controllers,” in *ACM SIGGRAPH 2025 Conference Papers*, 2025.
- [33] Z. Zhang, K. Wen, M. Xu, *et al.*, “Learning whole-body humanoid locomotion via motion generation and motion tracking,” *arXiv preprint arXiv:2604.17335*, 2026. [Online]. Available: <https://arxiv.org/abs/2604.17335>
- [34] E. Perez, F. Strub, H. de Vries, *et al.*, “FiLM: Visual reasoning with a general conditioning layer,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, 2018, pp. 3942–3951. [Online]. Available: <https://ojs.aaai.org/index.php/AAAI/article/view/11671>
- [35] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 6840–6851. [Online]. Available: <https://proceedings.neurips.cc/paper/2020/hash/4c5bfcfec8584af0d967f1ab10179ca4b-Abstract-Conference.html>